
```

def perturb_sa3(cities, itinerary, time, maxitin):
    neighborids1 = math.floor(np.random.rand() * (len(itinerary)))
    neighborids2 = math.floor(np.random.rand() * (len(itinerary)))
    global mindistance
    global minitinerary
    global minidx
    itinerary2 = itinerary.copy()
    randomdraw = np.random.rand()

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    small = min(neighborids1, neighborids2)
    big = max(neighborids1, neighborids2)
    if(randomdraw2 >= 0.55):
        itinerary2[small:big] = itinerary2[small:big][::-1]
    elif(randomdraw2 < 0.45):
        tempitin = itinerary[small:big]
        del(itinerary2[small:big])
        neighborids3 = math.floor(np.random.rand() * (len(itinerary)))
        for j in range(0, len(tempitin)):
            itinerary2.insert(neighborids3 + j, tempitin[j])
    else:
        itinerary2[neighborids1] = itinerary[neighborids2]
        itinerary2[neighborids2] = itinerary[neighborids1]

    temperature = 1 / (time / (maxitin / 10) + 1)

    distance1 = howfar(genlines(cities, itinerary))
    distance2 = howfar(genlines(cities, itinerary2))

    itinerarytoreturn = itinerary.copy()

    scale = 3.5
    if((distance2 > distance1 and (randomdraw < (math.exp(scale * (distance1 -
distance2)) * \temperature)) or (distance1 > distance2))):
        itinerarytoreturn = itinerary2.copy()

    reset = True
    resetthresh = 0.04
    if(reset and (time - minidx) > (maxitin * resetthresh)):
        itinerarytoreturn = minitinerary
        minidx = time

    if(howfar(genlines(cities, itinerarytoreturn)) < mindistance):
        mindistance = howfar(genlines(cities, itinerary2))

```